Ben Bubnick

Healthcare Data Consultant - Architecture, Governance, Integration

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Professional Summary

Healthcare data consultant with 10+ years translating clinical, payer, and revenue-cycle sources into governed analytics platforms. Led an integration spanning 70+ EHR sources, modeled 55 Snowflake/dbt tables, and validated deidentified data representing approximately 300 million patient lives.

Technical Skills

Healthcare data: Clinical and EHR data, payer claims and eligibility, revenue cycle, population health, HL7 v2/v3, FHIR, EDI/837, HIPAA
Architecture and delivery: Data modeling, source-to-target mapping, data quality, governance, metadata and lineage, deidentification, reconciliation, stakeholder requirements
Platforms and tools: SQL, Python, Snowflake, dbt, SQLMesh, AWS, Azure, Spark/Hadoop, Power BI

Professional Experience

HHAeXchange Oct. 2025 - Jun. 2026
Senior Data Engineer
Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, establishing reusable patterns for faster client onboarding.
- Conducted feasibility analysis of source-system structures and implementation constraints to shape integration scope, modeling strategy, and delivery approach.
- Translated ambiguous business requirements into documented mappings, acceptance criteria, onboarding standards, and 12 data-quality investigation runbooks.
- Created or materially redesigned 30 automated SQL/dbt data-quality tests covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.

Adonis Mar. 2025 - Jul. 2025
Senior Data Engineer
Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker, GitHub Actions

- Diagnosed undocumented source-key versioning and implemented composite business grains with surrogate keys across 10 charge and payment models, eliminating duplicate-key failures.
- Built, documented, validated, and maintained 54 SQLMesh models that delivered standardized revenue-cycle datasets for a provider whose network served 4,598 families across 113 locations and processed 86,388 annual authorizations.
- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
- Built a reusable SQLMesh/Snowflake ingestion and transformation framework using templated models, data contracts, CI checks, and promotion gates.

GrowData Analytics Sep. 2024 - Oct. 2025
Principal ML Engineer
Technologies: Python, Ruby, RubyLLM, RSpec, RuboCop, Tesseract OCR, LLM APIs, Docker, AWS, GitHub Actions

- Built and validated a preproduction document-to-data MVP that converted semi-structured invoices into auditable transaction records, achieving 95% field-level accuracy across a stratified 500-invoice test suite.
- Instrumented the MVP with automated tests, audit logging, retry controls, and confidence routing, achieving 88% straight-through processing and under 1% system failure.

Arkos Health Jun. 2023 - Jan. 2025
Senior Data Engineer
Technologies: AWS S3, Athena, Redshift, Glue, SNS, dbt, PySpark, Docker, Jupyter, Power BI, Tableau

- Architected and built a greenfield AWS data-lake foundation integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets; CI/CD automation and targeted model refactoring accelerated deployments and improved query performance.
- Integrated existing Jupyter-based analytics and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring, executive, and BI reporting.

Merative Jul. 2022 - Mar. 2023
Senior Data Scientist
Technologies: Microsoft Azure, Azure Data Factory, Blob Storage, Azure Functions, SQL Database, Synapse Analytics, Python, SQL, Azure DevOps

- Led HIPAA-compliant deidentification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.
- Engineered a seven-layer Python and SQL migration-acceptance suite with zero unresolved discrepancies at sign-off across tested record counts, partitions, schemas, nulls, duplicates, field values, aggregates, and business rules.

IBM Nov. 2021 - Jul. 2022
Senior Data Scientist
Technologies used across IBM roles: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, T-SQL, MySQL, PostgreSQL, PL/SQL, MQL, XPath, XQuery, Apache Pig, Jaql, JSONPath, Docker, Jenkins, Tableau, HL7

- Served as technical lead across electronic health records, insurance claims, scheduling and financial data, social determinants of health, and other relevant sources.
- Engineered a Spark, Hive, and Impala analytics framework powering 50+ prior-authorization lag-time and approval-bottleneck reports with daily operational and periodic regulatory and financial snapshots.

IBM Dec. 2020 - Nov. 2021
Senior Data Architect

- Served as a subject-matter expert to project data architects and delivery, analytics, and integration teams; directed configuration of the data-onboarding platform and supporting infrastructure.
- Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations, standardizing EDI batch schemas while enabling Python/Jupyter lineage tracing and end-to-end validation gates.
- Architected Spark and Hadoop pipelines integrating more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of measures and registries.

IBM Nov. 2017 - Dec. 2020
Data Architect

- Architected data onboarding for the Central New York Care Collaborative on IBM Watson Health's Explorys platform, integrating 75+ EHR, HIE, and other source systems across six counties; implemented provider-scoped patient filtering to enforce data-sharing rules and coordinated delivery across engineering teams, earning promotion to Senior Data Architect.
- Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing Jira-tracked validation rework by 25% and mapping rework by 45%.

IBM Sep. 2016 - Nov. 2017
Software Developer, Data Platform

- Engineered an HL7 v2/v3 parsing pipeline in Pig for approximately 30 internal users, replacing a Mirth-based workflow and eliminating $30,000 in annual vendor costs.

IBM Nov. 2015 - Sep. 2016
Data Scientist

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Built a Python, Ruby, and SQL extraction tool that eliminated unnecessary story points and reduced average engineering hours per client integration.

Explorys, an IBM Company Jun. 2015 - Nov. 2015
Data Scientist
Technologies: Hadoop/HDFS, Ruby, Pig, React, Node.js, JavaScript, SQL, Oracle, Jenkins

- Identified and extracted data from heterogeneous client systems; led technical requirements gathering, discovery, documentation, and implementation; and coordinated acquisition, installation, and configuration of VPN hardware supporting client integrations.

Education

M.S. Astrophysics, University of Cincinnati, Dec. 2010
B.F.A. Printmaking, Ohio University, Aug. 2004

Certifications

Certificate of Specialization in Data Science, Johns Hopkins University, 2015

IBM Data Science Professional Certificate, IBM, 2022